Reverse-Engineering Gene Regulatory Networks Using Dynamic Bayesian Networks

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Abstract

This study evaluates the efficacy of Dynamic Bayesian Networks (DBNs) in reconstructing the structure of Gene Regulatory Networks (GRNs) from discrete time-series expression data. Utilizing the exact structure-learning algorithm implemented in `BNFinder2`, we investigate the impact of network size, update dynamics (synchronous vs. asynchronous), temporal sampling frequency, and scoring criteria (Minimum Description Length vs. Bayesian-Dirichlet Equivalence) across both synthetic Random Boolean Networks ($N \in \{5, 8, 10, 13, 16\}$) and the 10-node Mammalian Cell Cycle biological network. Empirical results demonstrate that synchronous and transient pre-attractor trajectories consistently yield the highest precision (up to 1.00 on synthetic topologies and 0.82 on the biological network) and lowest Structural Hamming Distance (SHD = 24).

1 Introduction

Reconstructing causal regulatory interactions from time-series gene expression profiles is a central inverse problem in computational systems biology. Dynamic Bayesian Networks (DBNs) model temporal stochastic processes by restricting directed edges to point strictly forward in time ($t \to t+1$), thereby unrolling cyclic feedback loops into a bipartite Directed Acyclic Graph (DAG) and enabling exact polynomial-time parent set identification.

The study is structured in two experimental phases:

1. **Part I (Synthetic Topology Benchmark):** Systematic evaluation on Random Boolean Networks (RBNs) of increasing size ($N \in \{5, 8, 10, 13, 16\}$ nodes, in-degree $k \in \{1, 2, 3\}$) across four distinct dynamical regimes.

2. **Part II (Biological Network Reconstruction):** Validation on the 10-node Mammalian Cell Cycle network compiled from Boolean rules (`.bnet`) into exact truth tables.

2 Mathematical Formulation & Methodology

2.1 Dynamic Bayesian Network Structure Learning

Let $\mathbf{X}^{(t)} = (X_1^{(t)}, \ldots, X_n^{(t)}) \in \{0, 1\}^n$ denote the binary expression state vector at discrete time step t. Under the first-order Markov assumption, the transition probability factorizes over a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$:

$$P\left(\mathbf{X}^{(t+1)} \mid \mathbf{X}^{(t)}, \mathcal{G}, \Theta\right) = \prod_{i=1}^{n} P\left(X_i^{(t+1)} \mid \mathbf{Pa}_i^{(t)}, \theta_i\right) \tag{1}$$

where $\mathbf{Pa}_i^{(t)} \subseteq \mathbf{X}^{(t)}$ denotes the regulator parent set of node X_i. Given N observed transitions $\mathcal{D}$, we compare two decomposable scoring functions in `BNFinder2`:

- **Minimum Description Length (MDL):** Penalizes model complexity (parent set cardinality) to enforce parsimony and suppress spurious edges:

$$\mathrm{MDL}(X_i, \mathbf{Pa}_i : \mathcal{D}) = -\sum_{j=1}^{q_i} \sum_{k \in \{0,1\}} N_{ijk} \log_2 \left(\frac{N_{ijk}}{N_{ij}} \right) + \frac{\log_2 N}{2} q_i (r_i - 1) \tag{2}$$

  where $r_i = 2$, $q_i = 2^{|\mathbf{Pa}_i|}$, and N_{ijk} is the transition count from parent configuration j to child state k.

- **Bayesian-Dirichlet Equivalence (BDE):** Evaluates the marginal likelihood over a Dirichlet prior on transition parameters.

2.2 Simulation Regimes

To evaluate sensitivity to experimental conditions, four time-series regimes were simulated:

- **Sync Base (`sync_base`):** Deterministic simultaneous updates ($T = 100, \Delta t = 1$), representing noise-free trajectories.
- **Async (`async`):** Stochastic single-node updates per step ($T = 200, \Delta t = 1$), modeling biological update stochasticity.
- **Sampled (`sampled`):** Synchronous updates sub-sampled every 3rd step ($T = 150, \Delta t = 3$), modeling low temporal resolution.
- **Transient (`transient`):** Short synchronous bursts ($T = 20, \Delta t = 1$), capturing dynamic transitions prior to attractor convergence.

3 Part I: Synthetic Network Analysis

Table 1 and Figures 1–2b summarize reconstruction performance across network sizes $N \in \{5, 8, 10, 13, 16\}$ (fixed random seed 42).

- **Topology & Attractor Sensitivity:** Reconstruction accuracy depends heavily on attractor basin structure rather than node count alone. For $N = 5$, deterministic synchronous trajectories (`sync_base` and `transient`) rapidly converged to a fixed point, providing insufficient state transitions for MDL/BDE ($F_1 = 0.00$), whereas stochastic `async` perturbations escaped the attractor and achieved $F_1 = 0.29$ (Precision $= 0.67$). Conversely, at $N = 10$, `sync_base` and `transient` achieved strong recovery ($F_1 = 0.41$, Precision $= 0.86$ and $F_1 = 0.34$, Precision $= 0.71$, respectively).
- **Precision vs. Recall Bottleneck:** Across medium-to-large networks ($N \in \{10, 13, 16\}$), `sync_base` and `transient` regimes maintain high precision (up to 1.00 at $N = 13$), whereas recall drops sharply ($0.04 - 0.27$) as exponentially growing parent state combinations remain unobserved in a single trajectory.
- **Degradation under Sub-Sampling:** Recording every 3rd step (`sampled`) completely destroys the first-order Markov signal for $N \in \{5, 8, 10, 13\}$ ($F_1 = 0.00$) and yields only $F_1 = 0.06$ at $N = 16$.

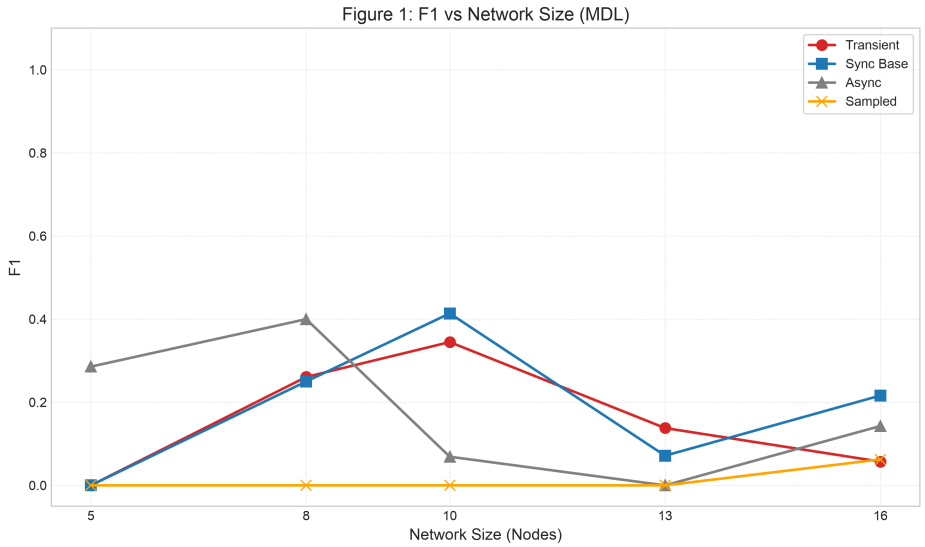
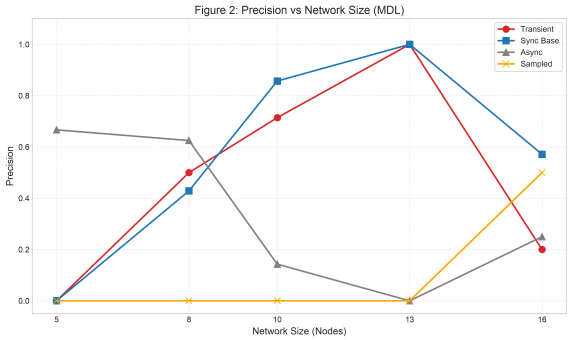
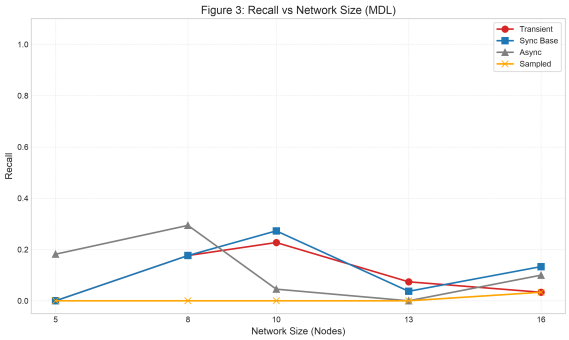


Figure 1: Part I: Reconstruction F_1 Score vs. Network Size under MDL scoring.



(a) Precision vs. Network Size (MDL)



(b) Recall vs. Network Size (MDL)

Figure 2: Decomposition of edge recovery into Precision and Recall across synthetic RBNs.

Table 1: Part I Empirical Results: Synthetic Random Boolean Networks ($N \in \{5, 8, 10, 13, 16\}$).

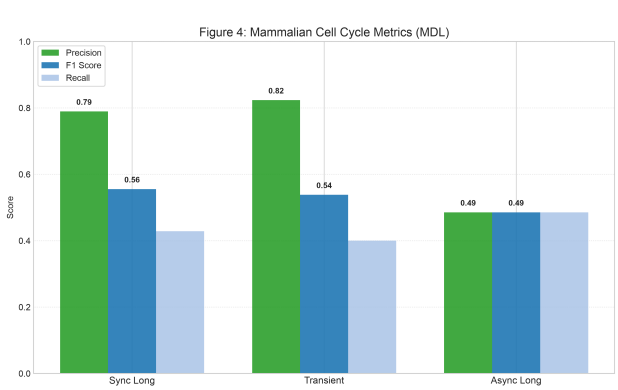
Dataset	Meth	Prec	Rec	F1	SHD	Dataset	Meth	Prec	Rec	F1	SH
data_5_sync_base	MDL	0.00	0.00	0.00	11	data_10_sampled	MDL	0.00	0.00	0.00	2
data_5_sync_base	BDE	0.00	0.00	0.00	11	data_10_sampled	BDE	0.00	0.00	0.00	2
data_5_async	MDL	0.67	0.18	0.29	10	data_10_transient	MDL	0.71	0.23	0.34	1
data_5_async	BDE	0.50	0.09	0.15	11	data_10_transient	BDE	0.43	0.14	0.21	2
data_5_sampled	MDL	0.00	0.00	0.00	11	data_13_sync_base	MDL	1.00	0.04	0.07	2
data_5_sampled	BDE	0.00	0.00	0.00	11	data_13_sync_base	BDE	0.00	0.00	0.00	2
data_5_transient	MDL	0.00	0.00	0.00	11	data_13_async	MDL	0.00	0.00	0.00	3
data_5_transient	BDE	0.00	0.00	0.00	11	data_13_async	BDE	0.00	0.00	0.00	3
data_8_sync_base	MDL	0.43	0.18	0.25	18	data_13_sampled	MDL	0.00	0.00	0.00	2
data_8_sync_base	BDE	0.33	0.12	0.17	19	data_13_sampled	BDE	0.00	0.00	0.00	2
data_8_async	MDL	0.62	0.29	0.40	15	data_13_transient	MDL	1.00	0.07	0.14	2
data_8_async	BDE	0.71	0.29	0.42	14	data_13_transient	BDE	1.00	0.07	0.14	2
data_8_sampled	MDL	0.00	0.00	0.00	17	data_16_sync_base	MDL	0.57	0.13	0.22	2
data_8_sampled	BDE	0.00	0.00	0.00	17	data_16_sync_base	BDE	0.40	0.07	0.11	3
data_8_transient	MDL	0.50	0.18	0.26	17	data_16_async	MDL	0.25	0.10	0.14	3
data_8_transient	BDE	0.50	0.18	0.26	17	data_16_async	BDE	0.18	0.07	0.10	3
data_10_sync_base	MDL	0.86	0.27	0.41	17	data_16_sampled	MDL	0.50	0.03	0.06	3
data_10_sync_base	BDE	0.83	0.23	0.36	18	data_16_sampled	BDE	0.50	0.03	0.06	3
data_10_async	MDL	0.14	0.05	0.07	27	data_16_transient	MDL	0.20	0.03	0.06	3
data_10_async	BDE	0.17	0.05	0.07	26	data_16_transient	BDE	0.60	0.10	0.17	2

4 Part II: Mammalian Cell Cycle Reconstruction

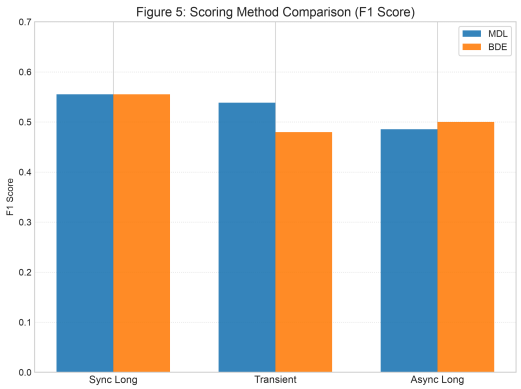
We applied the reconstruction pipeline to the 10-node Mammalian Cell Cycle network (35 ground-truth directed edges, including an implicit self-loop on input node `v_CycD`). For each regime, 5 independent trajectories were simulated and concatenated.

Table 2: Part II Empirical Results: 10-Node Mammalian Cell Cycle Network (35 Ground-Truth Edges).

Dataset Scenario	Method	Precision	Recall	F1 Score	Hamming Di
Mammalian_Cell_Cycle_sync_long	MDL	0.79	0.43	**0.56**	**24**
Mammalian_Cell_Cycle_sync_long	BDE	0.79	0.43	**0.56**	**24**
Mammalian_Cell_Cycle_transient_bursts	MDL	**0.82**	0.40	0.54	**24**
Mammalian_Cell_Cycle_transient_bursts	BDE	0.80	0.34	0.48	26
Mammalian_Cell_Cycle_async_long	MDL	0.49	0.49	0.49	36
Mammalian_Cell_Cycle_async_long	BDE	0.43	**0.60**	0.50	42



(a) MDL Performance across Regimes



(b) MDL vs. BDE Comparison (F_1)

Figure 3: Part II: Reconstruction performance on the Mammalian Cell Cycle network.

4.1 Discussion of Biological Reconstruction

1. **High Precision of Transient & Synchronous Regimes:** On the biological network, `transient_bursts` paired with MDL achieved the highest Precision (0.82) and tied with `sync_long` ($F_1 = 0.56, \text{Precision} = 0.79$) for the lowest Structural Hamming Distance (SHD = 24). Moreover, reconstruction on the 10-node biological network ($F_1 = 0.56$) substantially outperformed the 10-node synthetic RBN ($F_1 = 0.41$ for `sync_base`, 0.07 for `async`).

2. **Precision–Recall Trade-off in Asynchronous Data:** The `async_long` regime achieved the highest Recall (0.49 with MDL, 0.60 with BDE) by perturbing individual nodes out of synchronous lockstep, but suffered from false-positive edges (Precision $= 0.43 - 0.49$, SHD $= 36 - 42$).

3. **MDL vs. BDE:** MDL consistently matched or outperformed BDE in structural accuracy (SHD = 24 vs. 26 in transient; SHD = 36 vs. 42 in async) by penalizing spurious parent sets.

5 Conclusions

1. **Precision via Transient & Synchronous Perturbations:** Multi-replicate synchronous and transient trajectories yield the highest precision ($0.79 - 0.82$) and lowest structural edit distance (SHD = 24), making them reliable for biological hypothesis generation.

2. **Temporal Resolution Sensitivity:** Sub-sampling ($\Delta t = 3$) destroys first-order Markov transition signals ($F_1 = 0.00$ across $N \leq 13$); high sampling frequency over short bursts is strictly preferable to sparse long-horizon monitoring.

3. **Parsimony in Structure Scoring:** MDL regularization effectively suppresses false-positive edges compared to BDE, particularly under short transient bursts and stochastic asynchronous updates.